CST534 - Distributed Systems and Cloud Computing

CST 534 (Distributed Systems and Cloud Computing) Syllabus

Course Information

Course Details

- **Course Title:** Distributed Systems and Cloud Computing
- **Course Number:** CST 534
- **Meeting Location:** Physical
- **Units:** 4

Course Description

This course begins with an overview of computer architectures and operating systems as an introduction to modern computer systems. It then delves into distributed systems, focusing on the principles, design, and implementation of scalable, fault-tolerant, and high-performance systems. Topics include distributed algorithms, consensus mechanisms, consistency models, and fault tolerance, as well as computer systems concepts such as resource management, process scheduling, and memory architecture. The course also covers cloud computing architectures, including virtualization and containerization technologies. Students will analyze and develop distributed applications using cloud platforms, with an emphasis on data-intensive systems, resource management, and security.

Materials

- Course textbook: "Distributed Systems: Concepts and Design" by George Coulouris, Jean Dollimore, Tim Kindberg, and Gordon Blair
- Supplemental materials: "Designing Data-Intensive Applications" by Martin Kleppmann, cloud platform documentation (AWS/Azure/GCP), selected research papers on distributed systems, and current industry whitepapers on microservices and containerization

Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system - Cloud computing platform access (AWS, Azure, or Google Cloud Platform with educational credits) - Containerization tools (Docker, Kubernetes) for distributed application deployment - Distributed systems development and simulation environments

Course Objectives

At the end of this class, you should be able to:

1. Design and implement scalable, fault-tolerant distributed systems
2. Apply distributed algorithms and consensus mechanisms to solve complex problems
3. Utilize cloud computing platforms and containerization technologies effectively
4. Analyze and optimize resource management in data-intensive distributed applications

Grading

There are three groups of assignments, briefly described in the table below. **Getting below a 40% in any of these three groups will result in failing the class.**

Assignment Kind	Value
Projects	35%
Exams	35%
Participation	30%

Projects

Projects will take the form of distributed system design and implementation, cloud-native application development, containerized microservices deployment, consensus algorithm implementations, and fault-tolerant system prototypes. Students will build scalable applications using modern cloud platforms and analyze performance under various failure scenarios.

Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups. Therefore, attendance is required, as well as participation in all in-class activities. These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes: 1. Instructor identifies need for change 2. Revised syllabus section is prepared 3. Students are notified via Canvas and email 4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.